

System Architecture

HARDWARE

Pi Camera

12MP · CSI · 30fps

Raspberry Pi 5

8GB · Ubuntu 22.04

Pixhawk

PX4 · IMU · Baro

GPS Module

M8N · ±2m accuracy

4G LTE Dongle

USB · wide-area link

RC Receiver

2.4GHz · HW override

DRONE SOFTWARE — ROS 2

Frame Acquisition

OpenCV · BGR → RGB

YOLOv8n.pt

Person · Car

best.pt

Drone (custom)

Merge + Filter

conf ≥ 0.45 · class map

DeepSORT Tracker

Track ID · Kalman · IoU

GPS Fusion

Pixel → lat/lon estimate

IFF Node

Friendly / Unknown?

Friendly

Unknown

Discard

Alert Publisher

Mission Manager

FSM · waypoints · failsafe

CLOUD BACKEND

REST API

GET /alerts · /telemetry

PostgreSQL

JSONB · alerts · telemetry

FastAPI

WebSocket relay

GCS DASHBOARD

DroneContext

React state hub · WS client

Leaflet.js Map

Live position · path · markers

Alert Feed

Class · confidence · GPS

Telemetry Bar

Battery · Alt · Speed · Mode

Command Panel

RTH · Takeoff · Land · Waypoints